MINISTRY OF HIGHER EDUCATION
EGYPTIAN CHINESE UNIVERSITY

DEPARTMENT	: COMPUTERS AND INFORMATION SYSTEMS	DATE	:4/2025
COURSE NAME	: COMPUTER PROGRAMMING II	TIME ALLOWED	: 3WEEKS
COURSE CODE	: CSC-1200	TOTAL MARK	: 15
YEAR / LEVEL	: 1	NUMBER OF PAGES	:
COORDINATOR	: DR. AYMAN MOHAMED & DR DOAA MOHEY		:

PROJECT 6 **MARKS: [15GRADES+ 5 GRADES BONUS]**

Restaurant Management System Using C++ with GUI Interface

Project Idea:
As the food service industry grows increasingly reliant on digital systems for efficiency, a computerized restaurant management system becomes essential. This project aims to design a restaurant management system in C++ that can handle food ordering, table reservations, billing, and basic admin operations. The system will include a graphical user interface (GUI) and apply Object-Oriented Programming (OOP) principles to ensure modularity and maintainability.

Project Objectives:

1. Simulate a restaurant environment with order placement, billing, and table management.
2. Apply Object-Oriented Programming (OOP) concepts such as Encapsulation, Inheritance, Abstraction, and Polymorphism.
3. Enable role-based access (Admin and Customer).
4. Allow customers to view the menu, place orders, and reserve tables.
5. Allow the admin to manage menu items and view orders.
6. Use file storage to maintain menu and order data.

Tools and Technologies:

- **C++:**
  - **For high-performance image processing and real-time object detection.**
  - **Supports OOP principles (Encapsulation, Inheritance, Polymorphism, Abstraction).**

Programming Language: C++

GUI Library: GUI Interface for building the interface.
System Features:
Admin:

- Login to the system.
- Add new menu items (name, category, price).
- Edit or delete existing menu items.
- View all orders placed.
- View table reservations.

Customer:

- Register/Login.

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- Browse the menu by category.
- Place a food order and generate bill.
- Reserve a table for a selected date and time.
- View order history (optional).

Main Object-Oriented Programming (OOP) Concepts

1. **Encapsulation:**
   Menu, Orders, Reservations, and Users will be encapsulated in classes with their data and functions.
2. **Abstraction:**
   User interfaces and system operations (like placing an order) will be abstracted into user-friendly functions.
3. **Inheritance:**
   Admin and Customer classes will inherit from a base User class.
4. **Polymorphism:**
   Display or manage operations may behave differently for Admin vs Customer through function overriding.

Methodology:

1. Interactive GUI with buttons, input fields, and tables for order/menu/reservation display.
2. Structured user flow for both Admin and Customer roles.
3. Dynamic order processing system with real-time bill generation.
4. Reservation system with time-slot selection.
5. Admin dashboard for tracking and managing data.
6. File I/O operations to persist:
   a. Menu data.
   b. Orders and reservations.
7. User information and login credentials.

Project Applications:

- **User:** Base class with name, email, and password.
- **Admin:** Inherits from User and can manage menu/reservations.
- **Customer:** Inherits from User and can order/reserve.
- **MenuItem:** Contains name, category, price.
- **Order:** Contains ordered items, total amount, and timestamp.
- **Reservation:** Contains customer name, table number, date, and time.

Bonus:

- Text-based receipt generation and order logs.
- Table availability display in real time.
- Password hashing for secure login.
- Sorting or filtering menu items.
- Daily sales report for admin.